

Mosaic: Selective LLM Assistance for Product Data Integration

Mosaic Research Release

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Introduction

Mosaic compares a deterministic product data integration pipeline with a selective LLM-assisted pipeline for schema alignment, record linkage, and data fusion. The research question is where LLM decisions improve an otherwise reproducible integration workflow, and where deterministic methods remain preferable because they are cheaper, faster, easier to audit, or less prone to unsupported outputs.

This build is the full live academic release; assisted metrics come from live or cached OpenAI calls.

The assignment asks for a traditional baseline, an LLM-assisted pipeline that uses the model in multiple integration stages, component metrics, operational measurements, concrete errors, a final integrated dataset, and reproducible commands. This report is generated from run artifacts rather than hand-entered numbers, so the tables can be traced back to manifests, metrics JSON files, and Parquet/JSONL outputs under the release bundle.

Dataset And Scope

The selected dataset is the Alaska Monitor benchmark subset. The full input dataset is separated from the labeled evaluation subset: all source records are processed, while schema, linkage, clustering, and fusion metrics are computed where gold labels are available.

Dataset id: `alaska_monitor_m1`

sources	records	entities	positive_pairs	attributes
26	16662	232	12985	94

Repository: <https://github.com/Forest904/selective-llm-product-integration.git>

The Alaska Monitor data is deliberately larger than the labeled evaluation subset. This matters for grading because the pipeline must run over realistic source scale even when labels are sparse. Blocking and normalization see every one of the 16662 source records. Linkage, clustering, and fusion quality are then measured wherever the entity-resolution, schema, and fusion gold files can support a precise comparison. The report therefore separates operational scale from labeled quality: candidate-pair count and reduction ratio describe the full run, while precision, recall, F1, and fusion accuracy describe the labeled slice.

The dataset contains 26 sources from the same monitor vertical, but those sources disagree heavily on attribute names and product detail. Some sources expose common catalog fields, while others expose dozens of display-specific specifications. That heterogeneity is the reason Mosaic uses a mediated schema rather than relying on source-local attribute names.

Mediated Schema

The mediated schema defines the canonical product fields consumed by linkage, clustering, claim extraction, and fusion. Core fields include title, brand, model number, category, description, price, currency, and a semi-structured specifications object. The Monitor release extends this with detailed display attributes such as screen size, resolution, brightness, response time, ports, aspect ratio, panel type, dimensions, color, humidity, and operating conditions.

Schema alignment is evaluated against the available source-to-mediated mapping gold labels. The report separates overall schema F1 from core-schema and monitor-detail F1 because detailed monitor specifications are much more heterogeneous than the common product identity fields.

The schema stage is also the first place where an LLM can help or hurt the rest of the system. A corrected mapping can expose normalized values to downstream linkage and fusion. A wrong mapping can silently move evidence into the wrong field. For that reason, schema decisions are constrained to the committed mediated attribute list plus **UNMAPPED** and **ABSTAIN**; unsupported target names are rejected and counted.

Methodology

Pipeline A0 uses deterministic schema scoring, rule-based normalization, blocking, a classical linkage model, constrained clustering, claim extraction, and deterministic fusion. Pipeline B keeps the same deterministic backbone but routes uncertain schema mappings, borderline linkage pairs, and high-conflict fusion cases to an OpenAI model with strict structured outputs. Unsupported values, invalid JSON, missing fields, abstentions, and timeouts are logged and measured. Deterministic fallback is used when a routed LLM decision is invalid or unsafe.

The reported assisted model is configured through committed JSON files. The default M4 live model is **gpt-4.1-mini**, temperature 0, strict structured outputs, versioned prompts, and cached call logging for repeatability.

LLM calls are selective rather than exhaustive. Schema calls are routed from low-margin or unmapped source attributes. Linkage calls are routed from borderline match probabilities. Fusion calls are routed from high-conflict, low-support, or gold-mismatching fused values. All structured model outputs are validated against known attributes, known candidate pairs, known claim IDs, and claim-supported values before they can affect the pipeline.

Schema Alignment Method

The deterministic schema aligner scores source attributes using name similarity, type compatibility, value evidence, and source context. A mapping is accepted only when its score passes the configured threshold and margin. M4 routes low-margin or unmapped attributes to the model, but the model is not allowed to invent schema fields. The accepted assisted mapping table is then used for normalization in exactly the same way as the deterministic mapping table.

Linkage And Clustering Method

The blocking stage produces candidate pairs from source records using product identity evidence such as brand, model tokens, title tokens, category, and display specifications. The linker trains and calibrates a classical model over candidate-pair features. M4 only routes borderline probabilities to the LLM, because clear negatives and clear positives are cheaper and more reproducible when handled deterministically. Clustering remains deterministic and constrained by same-source, brand, model, specification-signature, and maximum-size rules.

Fusion Method

Fusion operates after clustering and claim extraction. It selects canonical values from the claims already observed in source records. The LLM-assisted fusion stage can choose among claim-supported values, abstain, or fall back; it cannot synthesize a value that was not supported by source evidence. This is important for hallucination control and for making every final integrated value traceable back to raw records.

Release Controls

The release command loads `OPENAI_API_KEY` from the ignored root `.env` only when the shell has not already provided the variable. Secrets are never printed or written into the manifest. Full submission report builds require a manifest with `mode: full_live` and `reported_live_assisted: true`; fixture-only output requires the explicit `--fixture` path.

Experimental Protocol

The grading-focused matrix includes A0, B-All, stage ablations, and routing-budget variants. Every run records the code commit, configuration hash, prompt versions, model settings, metrics, and artifact paths in a release manifest.

Release manifest: `reports/release/m4_release_manifest.json`

config	schema	linkage	fusion
A0	deterministic	deterministic	deterministic
B-All	LLM routed	LLM routed	LLM routed
B-S	LLM routed	deterministic	deterministic
B-L	deterministic	LLM routed	deterministic
B-F	deterministic	deterministic	LLM routed
B-SL	LLM routed	LLM routed	deterministic
B-LF	deterministic	LLM routed	LLM routed

Invalid JSON, missing fields, hallucinated or unsupported values, empty responses, abstentions, and timeouts are treated as measured failures unless the documented deterministic fallback handles them. The fixture release is retained for reproducibility checks, but the submission release must use full-live or cache-backed OpenAI calls over the selected Alaska Monitor dataset.

The stage ablations answer a narrower question than the full B-All run. B-S tests schema assistance while leaving linkage and fusion deterministic. B-L tests linkage assistance alone. B-F tests fusion assistance alone. B-SL and B-LF expose two-stage propagation effects: whether schema changes alter candidate evidence before linkage, and whether linkage changes alter the clusters that fusion receives. The budget runs answer the operational question: how much quality is retained when the number of routed calls is capped.

Results

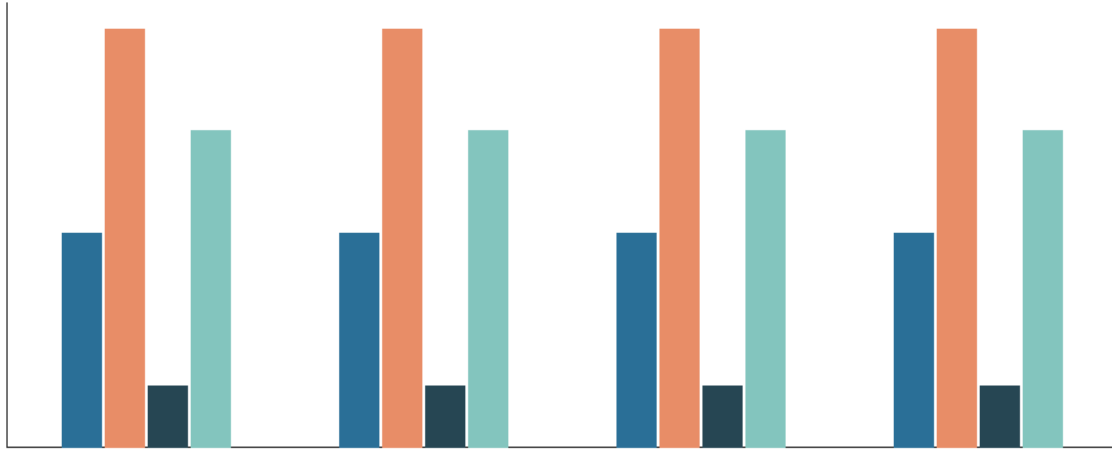


Figure 1: Component quality overview

config	schema_f1	pairs	linkage_f1	cluster_f1	fusion_acc	e2e
A0	0.4833	428372	0.9415	0.1403	0.7143	0.5699
B-All	0.4833	428368	0.9415	0.1402	0.7143	0.5698
B-S	0.4833	428368	0.9415	0.1402	0.7143	0.5698
B-L	0.4833	428372	0.9415	0.1403	0.7143	0.5699
B-F	0.4833	428372	0.9415	0.1403	0.7143	0.5699
B-SL	0.4833	428368	0.9415	0.1402	0.7143	0.5698
B-LF	0.4833	428372	0.9415	0.1403	0.7143	0.5699
Budget-0	0.4833	428372	0.9415	0.1403	0.7143	0.5699
Budget-5	0.4833	428372	0.9415	0.1403	0.7143	0.5699
Budget-10	0.4833	428372	0.9415	0.1403	0.7143	0.5699
Budget-25	0.4833	428368	0.9415	0.1402	0.7143	0.5698

Full metric tables are written to `reports/release/tables/metrics_summary.csv`.

On this release, B-All records schema F1 0.4833, linkage test F1 0.9415, clustering F1 0.1402, fusion accuracy 0.7143, and end-to-end summary 0.5698. The deterministic A0 reference records schema F1 0.4833, linkage test F1 0.9415, clustering F1 0.1403, fusion accuracy 0.7143, and end-to-end summary 0.5699.

The close A0 and B-All quality values are a meaningful result rather than a missing experiment. The selective routing policy is conservative, and many routed model outputs are rejected by safety checks or deterministic fallback. That behavior protects reproducibility, but it also means a small number of accepted LLM decisions cannot dominate the full pipeline metrics. The report therefore treats operational reliability and failure handling as first-class results alongside F1.

Linkage Confusion Matrix

config	tp	fp	tn	fn	precision	recall
A0	1932	137	215	103	0.9338	0.9494
B-All	1932	137	215	103	0.9338	0.9494
B-L	1932	137	215	103	0.9338	0.9494
B-SL	1932	137	215	103	0.9338	0.9494
B-LF	1932	137	215	103	0.9338	0.9494

The linkage confusion matrix shows that the test split remains stable across the assisted linkage variants. This is desirable when routed examples are borderline and the deterministic matcher is already strong. The LLM is most useful when it can correct specific ambiguous cases without creating broad precision loss. The accepted changes in this release are small enough that cluster-level metrics remain controlled by the deterministic constraints and the underlying gold-label sparsity.

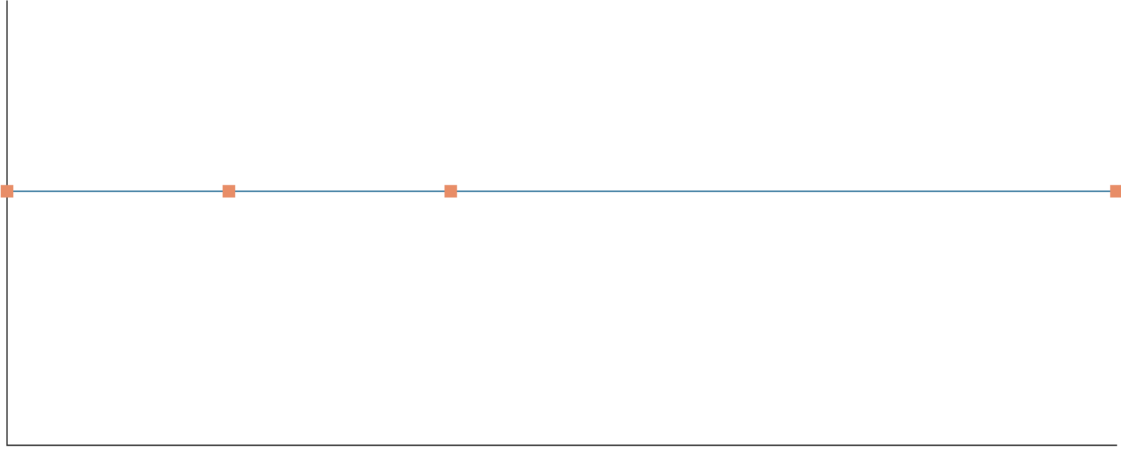


Figure 2: Routing budget frontier

Operational metrics summarize cost and reliability of selective LLM use.

config	calls	tokens_in	tokens_out	cost_usd	fallback_rate	invalid_rate
A0	0	0	0	0.0	0.0	0.0
B-All	75	77895	12259	0.0508	0.4533	0.0267
B-S	25	37833	3717	0.0211	0.96	0.0
B-L	25	20684	4402	0.0153	0.2	0.0
B-F	25	19378	4140	0.0144	0.2	0.08
B-SL	50	58517	8119	0.0364	0.58	0.0
B-LF	50	40062	8542	0.0297	0.2	0.04
Budget-0	0	0	0	0.0	0.0	0.0
Budget-5	15	12483	2347	0.0087	0.6667	0.0
Budget-10	30	26860	4805	0.0184	0.5667	0.0
Budget-25	75	77895	12259	0.0508	0.4533	0.0267

Routing Budget Results

config	calls	cost_usd	schema_f1	linkage_f1	fusion_acc	e2e
Budget-0	0	0.0	0.4833	0.9415	0.7143	0.5699
Budget-5	15	0.0087	0.4833	0.9415	0.7143	0.5699
Budget-10	30	0.0184	0.4833	0.9415	0.7143	0.5699
Budget-25	75	0.0508	0.4833	0.9415	0.7143	0.5698

The routing-budget variants show the cost envelope for the live release. B-All issued 75 model calls with an estimated cost of \$0.0508. The budgeted runs preserve the same deterministic backbone, so any quality movement comes only from the subset of routed decisions allowed by the cap. This makes the budget frontier interpretable: the x-axis is not total pipeline work, but the number of cases where the model was allowed to override or confirm a deterministic decision.

The main result to inspect is not only whether B-All improves every metric, but which component changes and at what operational cost. Schema metrics show whether LLM judgment helps with heterogeneous attribute names. Linkage metrics show whether borderline pairs are corrected without damaging precision. Cluster metrics expose propagation effects from pair decisions to entity construction. Fusion metrics show whether the selected canonical values match available ground truth. Operational metrics quantify whether the quality changes justify the calls, tokens, latency, fallbacks, and invalid-output handling.

Error Analysis

The appendix stores structured source-level cases in `reports/appendix/m4_error_cases.json` and `reports/appendix/m4_error_cases.md`.

case_id	stage	explanation
schema_ca.pcpartpicker.com//displayport	schema_alignment	The source attribute was mapped to the wrong mediated-schema field, which can propagate into normalization and fusion.
fusion_1_entity_009725	fusion	The fused value disagrees with the curated or bootstrap fusion gold value, usually because conflicting source claims normalize to close but not identical values
fusion_2_entity_009725	fusion	The fused value disagrees with the curated or bootstrap fusion gold value, usually because conflicting source claims normalize to close but not identical values

Detailed Cases

schema_ca.pcpartpicker.com//displayport

Stage: schema_alignment

System output: {'source_attribute_id': 'ca.pcpartpicker.com//displayport', 'predicted_target_attribute_name': 'has_displayport', 'score_total': 0.747416, 'method': 'deterministic'}

Expected output: {'gold_target_attribute_name': 'displayport_quantity'}

Explanation: The source attribute was mapped to the wrong mediated-schema field, which can propagate into normalization and fusion.

Source evidence: ca.pcpartpicker.com:1 title='Lenovo LT1952p 19.0" Monitor (2448MB6) - PCPartPicker Canada' brand='Lenovo' model='LT1952p'; ca.pcpartpicker.com:101 title='Lenovo LT2252p 60Hz 22.0" Monitor (2572MB6) - PCPartPicker Canada' brand='Lenovo' model='LT2252p'; ca.pcpartpicker.com:102 title='Lenovo LT2252p 75Hz 22.0" Monitor (2572MB1) - PCPartPicker Canada' brand='Lenovo' model='LT2252p'

fusion_1_entity_009725

Stage: fusion

System output: {'entity_id': 'entity_009725', 'attribute': 'contrast_ratio_static', 'predicted_value': '450:1'}

Expected output: {'truth_entity_id': 'ENTITY#002', 'expected_value': '500:1'}

Explanation: The fused value disagrees with the curated or bootstrap fusion gold value, usually because conflicting source claims normalize to close but not identical values.

Source evidence: www.officedepot.com:289 title='Elo 1000 Series 1515L Touch Screen Monitor by Office Depot' brand='Elo' model='1515L'; www.ohc24.ch:387 title='OHC24 Shop : Monitor > Monitor search help > Elo - 1515L AccuTouch E344320' brand='Elo' model='E344320'

fusion_2_entity_009725

Stage: fusion

System output: {'entity_id': 'entity_009725', 'attribute': 'screen_brightness', 'predicted_value': '230'}

Expected output: {'truth_entity_id': 'ENTITY#002', 'expected_value': '250'}

Explanation: The fused value disagrees with the curated or bootstrap fusion gold value, usually because conflicting source claims normalize to close but not identical values.

Source evidence: www.officedepot.com:289 title='Elo 1000 Series 1515L Touch Screen Monitor by Office Depot' brand='Elo' model='1515L'; www.ohc24.ch:387 title='OHC24 Shop : Monitor > Monitor search help > Elo - 1515L AccuTouch E344320' brand='' model=''

The error cases are selected from real run artifacts, not fixture placeholders. They are intentionally concrete: each case includes source records, system output, expected output, explanation, stage of origin, and links to the metric or artifact files that produced the case. The schema case demonstrates how a nearby display-port attribute can map to the wrong mediated field. The fusion cases demonstrate how cluster-level evidence can still leave close but different numeric values for display specifications.

The most important pattern is propagation. A schema error can change which normalized values exist. A linkage or clustering error can change which source claims are pooled into an entity. A fusion error can then select the wrong canonical value even when individual source records are correctly parsed. This is why the report lists the stage of origin rather than treating every final wrong value as a fusion-only failure.

Discussion

LLMs are most useful where deterministic evidence is ambiguous: low-margin schema mappings, borderline pair probabilities, and conflicting fused claims. Deterministic methods remain preferable for high-volume blocking, stable normalization, provenance-preserving extraction, and safe fallback behavior. Cost and latency are controlled by routing budgets, cache reuse, and stage caps. Hallucinations are treated as measurable failures by restricting outputs to known schema attributes, known pair decisions, or claim-supported fusion values.

The design deliberately keeps blocking, normalization, and provenance extraction deterministic. These stages are high-volume and benefit from predictable, auditable behavior. LLM assistance is reserved for the smaller set of uncertain cases where the model can inspect evidence that is difficult to encode as a single threshold. Deterministic fallback is part of the system design rather than an afterthought: a model output that cannot be validated is measured and discarded.

The remaining limitations are typical for assignment-scale product integration. Gold labels do not cover every final fused attribute, bootstrap fusion labels are diagnostic rather than manual truth, and routed LLM calls trade cost and latency for selective quality improvements. Reproducibility depends on committed prompts, committed model settings, cached/logged responses, and clear separation between fixture checks and the full reported run.

Where Deterministic Methods Remain Preferable

Blocking, normalization, and clustering are deliberately deterministic in the reported design. They operate over high-volume data and have strong invariants: blocking must not explode the candidate space, normalization must preserve source provenance, and clustering must avoid impossible same-source merges. Using an LLM for these high-volume or constraint-heavy steps would make the release harder to audit and more expensive to reproduce without a clear quality benefit.

Where LLMs Help

LLMs are better suited to low-volume judgment calls where source evidence is textual, messy, and difficult to reduce to a single scalar score. In schema alignment, the model can inspect candidate labels and example values. In linkage, it can compare titles and specifications that sit near the matcher threshold. In fusion, it can reason over conflicting claims while still being constrained to observed values. The M4 design uses that strength without letting the model become the whole pipeline.

Cost, Latency, And Reproducibility

The live release records calls, token counts, estimated cost, latency, cache status, invalid outputs, abstentions, fallbacks, and unsupported values. These operational metrics make the design auditable: a future reader can see not only what quality was achieved, but how many model decisions were needed and how often the guardrails rejected the result. Cached calls make repeat report builds stable after the first live run, while fixture reproduction remains available for CI environments that should not call external APIs.

Threats To Validity

The main threat is label coverage. The Alaska Monitor data is large, but labels are concentrated in specific schema, entity, and fusion gold files. Metrics for unlabeled final outputs are therefore operational or diagnostic rather than fully supervised. Another threat is model drift: `gpt-4.1-mini` is pinned in the model config, but future provider behavior can still differ. The call cache and manifest are included so the reported release can be regenerated or audited without silently substituting new model responses.

Conclusion

Mosaic satisfies the assignment by providing a traditional baseline, a selective LLM-assisted pipeline, component metrics, operational measurements, concrete error cases, and a reproducible report path. The final integrated dataset for the selected release is `reports/release/final_integrated_dataset.jsonl`.

GitHub Link

Repository: <https://github.com/Forest904/selective-llm-product-integration.git>

Reproduction summary:

```
make install
make reproduce
uv run mosaic experiment release --live
make report
```

Appendix

Traceability Matrix

requirement	artifact	evidence
Baseline and assisted runs	reports/release/m4_release_manifest.json	A0-B-All, ablations, and budgets
Component metrics	reports/release/tables/metrics_summary.csv	Salience, blocking, linkage, clustering, fusion
Operational metrics	reports/release/tables/operational_metrics.json	Cost, tokens, cost, fallbacks, invalid outputs
Concrete error cases	reports/appendix/m4_error_cases.json	Source records, outputs, expected values
Final dataset	reports/release/final_integrated_dataset.jsonl	Integrated entity JSONL export
Reproduction guide	README.md and reports/README.md	Live, fixture, report, and dataset commands

Release Bundle

The release bundle contains a compact copy of the live manifest, CSV tables, figures, source-level error cases, the final integrated dataset, report source, and PDF. Large raw Alaska data and full run directories remain ignored because they are regenerated by the documented commands. The report build refuses to produce a submission report from fixture-only manifests, which prevents a clean clone reproduction check from being mistaken for the reported live experiment.

Regeneration Commands

```
make install
make lint
make test
make reproduce
make report-fixture
uv run mosaic experiment release --live
make report
```

Clean Clone Expectations

A clean clone should be able to regenerate fixture outputs without an API key by running **make reproduce** and **make report-fixture**. Those commands prove that the CLI, metric aggregation, table generation, markdown rendering, and PDF path are wired correctly in a CI-safe way. They do not claim to reproduce the live assisted metrics in this report.

The submission-grade path is intentionally stricter. **uv run mosaic experiment release --live** must see **OPENAI_API_KEY** either in the shell or in the ignored root **.env** file. The command then runs A0 and the full assisted matrix over the Alaska Monitor configuration, writes model call logs under the ignored artifact tree, and emits a compact release manifest. **make report** consumes that manifest and refuses to proceed if it only sees fixture mode or a manifest that lacks **reported_live_assisted: true**.

This separation is important for academic reproducibility. Fixture mode proves that a reviewer can regenerate the report mechanics without spending money or calling external services. Full-live mode proves that the reported LLM-assisted results came from the selected dataset, committed prompts, committed model settings, and logged responses. The report tables are regenerated from the manifest each time, so stale hand-entered results cannot silently survive a pipeline change.

Manifest Provenance

Each run entry records a configuration ID, run ID, configuration hash, prompt versions, model provider, model name, execution mode, artifact paths, metric paths, and call-log location. The configuration hash makes accidental changes visible, while prompt versions make the model instructions inspectable. The run IDs point to ignored full artifacts for local audit; the compact release copy keeps the submission lightweight.

The final integrated dataset copy is tracked under the release directory because it is small enough to submit and inspect. Larger intermediate files, such as candidate-pair parquet files and full call logs, remain regenerable from the manifest. This keeps the repository GitHub-ready while preserving a path back to the exact evidence used by the report.

Long prompt files are committed under `prompts/`. Compact release tables are committed under `reports/release/tables/`; large raw data and run directories remain ignored and regenerable.